



REF : CSHK/DQT/QUA/2019/056

關於加強深開挖臨時支撐工程(ELS)質量監控工作的通知

各有關地盤：

質量技術部近期派員到各有關地盤進行質量檢查，發現部份地盤仍未嚴格落實公司《深開挖質量檢查及監控計劃》的要求。

現重申各地盤必須嚴格落實公司《深開挖質量檢查及監控計劃》的要求(詳見附件)，包括：

1. 委任合資格的深開挖臨時支撐工程(ELS)的監控人員，包括 IR1 及 IR2 人員；
2. 嚴格落實 IR1 及 IR2 人員的深開挖臨時工程各施工階段的質量監控職責；
3. 按照批准的圖則及方案做好各階段開挖及臨時支撐裝拆工程；
4. 未經批准不可有任何超挖情況；
5. 保留所需記錄。

請各有關地盤全面覆核及做好自身在建的深開挖各項施工工序及保存好所需記錄，質量技術部將派員跟進檢查。

如有任何問題請與質量技術部潘仲光先生聯絡，電話:28233620 或電郵: chungkwong_poon@cohl.com。

附件:《深開挖質量檢查及監控計劃》適用於 $\geq 4.5\text{m}$ 深開挖工程
(Ref: CSHK//DQT/QUA/2019/004)
(註: $\leq 4.5\text{m}$ 之開挖工程需按照業主批准施工圖則及方案進行)

二〇一九年十月十六日



抄報： 孔總、黃副總、張副總、劉副總、陳副總、宋助總
抄送： 房屋工程部、土木工程部、基礎工程部



中國建築工程(香港)有限公司
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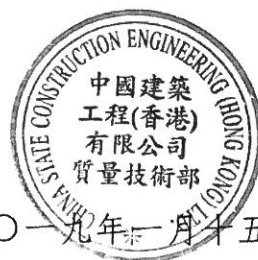
REF: CSHK/DQT/QUA/2019/004

關於印發《深開挖質量檢查及監控計劃》(英文版)的通知

各有關工程地盤:

為進一步加強各有關工程地盤深開挖工序的質量管理及監控工作,
現印發《深開挖質量檢查及監控計劃》(英文版), 詳見附件。

請認真遵照執行。



二〇一九年一月十五日

抄報: 孔總、黃副總、張副總、劉副總、陳副總、秦助總
抄送: 房屋部、土木部、基礎部



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Quality Inspection/Supervision Plan for Deep Excavation



**Standard for Execution of Deep Excavation Procedure
(in accordance with the Approval Drawings and Construction Method)**

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1. Purpose

As part of the Construction Process Control in the Management Quality Plan of the Project, this Quality Inspection/Supervision Plan (IP) describes the Witness and Hold Point inspection requirements in any deep excavation works involving multi-level strutting/tie-back lateral support system. It establishes the roles and responsibilities of the parties involved in the inspection procedures after commencement of the excavation to ensure compliance with procedural and safety criteria established in the design/construction documents. This IP is to be read in conjunction with the Code of Practice for Site Supervision 2009 (COP 2009) published by the Buildings Department. The duties established in this IP shall be over and above the duties to the relevant personnels as required in the above-mentioned Code of Practice in Building works.

This IP shall be maintained separately as a live document and is an additional supervision provision by the Registered Contractor (RC) to the Site Supervision Plan(SSP) required under the COP2009.

In house quality control team shall be responsible for the independent audit on the implementation of this IP.

2. Site Supervision Team

The following site supervision personnel will be involved for quality inspections for deep excavation:

- i. IR1: Designated Inspection Team Representative (a TCP at level T3 or above)
- ii. IR2: Designated Inspection Team Representative (a TCP at level T4 or above, or Site Agent / Project Manager / Project Director) been authorized to establish /execute Hold Point as defined in this IP

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- iii. ELS Designer: The design engineer of the temporary excavation works

lateral support system. He shall be the Registered Geotechnical Engineer (RGE) or Registered Structural Engineer (RSE) of the ELS in Building works, or the Consultant in Public Works Projects (Engineers' Design) or the Consultant employed by the Contractor in Design and Build Projects or the in-house Design Engineer employed by the Contractor as the case may be.

- iv. ICE: Independent Checking Engineer for the certification of the temporary work design and the construction certificate

Table of Site Supervision Team and Qualification:

Name of Supervision Team	Qualification Requirement	Equivalent to
IR1 - Designated Inspection Team Representative	TCP at level T3 or above	Table 2 of Technical Memorandum for Supervision Plans 2009
IR2 - Designated Inspection Team Representative	TCP at level T4 or above	Table 2 of Technical Memorandum for COP2009
	Site Agent/Project Manager/Project Director	Refer to the Project Organization Chart
ELS Designer		
● RGE/RSE	Section 3 of the Buildings Ordinance (BO)	Section 3 of the Buildings Ordinance (BO)
● Engineers' Design for Public Works Project	Category I supervision personnel	APPENDIX 7.47 of Project Administration Handbook for Civil Engineering Works
● Consultant	TCP at level T4/T5	Table 2 of Technical Memorandum for COP2009
● In-house Design Engineer	TCP at level T4/T5	Table 2 of Technical Memorandum for COP2009
ICE	Independent Checking Engineer	As per the contract requirement

When a Designated Inspection Team Representative is on leave or unavailable, replacement of equivalent qualification or better shall be assigned to take up the corresponding duties ascribed to him/her in this IP.



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3. Definitions

- i. Deep Excavation: As defined in COP 2009 clause 7.1(b) for excavation with depth exceeding 4.5 m.
- ii. Inspection: Examination carried out by IR1/IR2 via visual check of executed work through measurement or monitoring devices, to ensure the work is constructed in strict compliance with the dimensional, structural, procedural and safety requirements established in the design/construction documents.
- iv. Witness Point: A point defined in this procedure and identified during the excavation work in which the IR1 carries out daily Inspection and confirms compliance of conditions as described in the IP. Work may proceed during this process, unless the result of the Inspection dictates otherwise leading to the establishment of a Hold Point.
- v. “Alert” or “Alarm” Level : As defined in PNAP App-137.
- vi. Hold Point: A point at which the IR2 is required to establish after receipt of report by IR1 that either non-compliance of design conditions has been observed or a particular stage of excavation has been reached. Excavation shall not proceed until the IR2 is satisfied with the situation as described in procedure item 5 (iii) or (iv) below has been rectified or completed respectively and passes the Hold Point.

4. Inspection Procedure before Commencement of Excavation

- i. Before commencement of the excavation work, the designated IR2 shall ensure the preparation works are completed as per the approval plans/construction drawings and method statements. The checklist or Preparation Works Proforma shall be endorsed by IR2. The Preparation Works Proforma (Form A) is shown in **Appendix 2**.



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5. Inspection Procedure after Commencement of Excavation

- i. Upon commencement of the excavation work, the IR1 shall carry out routine Inspection on daily basis that the works comply with the approved plans, method statements, precautionary and protective measures. Particular attention shall be paid to the daily monitoring results recorded at monitoring stations as established according to the approved plans/construction drawings. The IR1 shall confirm that these results are all within the “Alert” limits as shown in the design drawings/ specification in a daily Witness Point Proforma (Form B1) as shown in **Appendix 3**.
- ii. The IR1 shall also check daily that the excavation level for any particular stage is not exceeded at any one time. He/she shall base his/her checking by measurement against reference level markers established by the project land surveyors at suitable locations on the temporary works. He/she shall record the excavation level(s) as measured in the daily Witness Point Proforma (Form B1) as shown in **Appendix 3**.
- iii. In the event that the IR1 observes during his/her routine Inspection any of the following situations:
 - A. the approved plan and method statement not being followed, or
 - B. the excavation level has exceeded the designed level for any stage, or
 - C. any of the monitoring results at monitoring check points has exceeded the “Alert” level, or
 - D. site constraints/situations which prevent the installation of lateral support system from conforming with the design and requires some form of alteration/modification to the dimension, orientation, level to the temporary works, or
 - E. abnormal movement, deformation or tilting in the temporary works, or



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- F. ground subsidence or potholes within or near the temporary works, or
- G. profuse quantity of seepage water seeps into the temporary works from any sources (such as water pipe burst etc), or
- H. report received of adjacent gas leak, damage to adjacent utilities or properties that may possibly be due to the excavation work, or
- I. accident occurred on excavation site involving the need of rescue operations, or
- J. major deviations in geological conditions stated in the approval plan

he/she shall immediately advise IR2 / ELS Designer in Non-Conformity Report Proforma Form B2 as shown in **Appendix 4**. The IR2 shall carry out an Inspection himself/herself and may, if required, after consulting the RGE / RSE / ELS Designer as the case may be, establish a Hold Point in a Hold Point Proforma Form C (as shown in Appendix 5) at this juncture. Excavation shall be temporarily suspended until IR2 / ELS Designer / or ICE has reviewed the situation and established remedial measures where necessary and passed this Hold Point after the remedial measures (if required) are completed to his/her satisfaction and the Hold Point Proforma Form C as shown in **Appendix 5** is signed off.

If IR2 considers the situation may impose imminent danger to the public, IR2 shall notify RGE / RSE / or ELS Designer and the site supervision team established under the SSP immediately so as to invoke the necessary follow-up action to notify Buildings Department or other government departments as the case may be.



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- iv. A Hold Point is also established when the required level for a particular stage is reached and strutting/tie-back is to be installed. The IR1 shall inform IR2 / ELS Designer / or ICE who shall carry out such Inspection as required to allow the lateral support system to proceed. No further excavation shall proceed until IR2 / ELS Designer / or ICE satisfies himself/herself that the lateral support system has been installed according to the approved plans/construction drawings and passes the Hold Point by signing off the Hold Point Proforma Form C.
- v. In the event that a non-compliance record is received from the client/project engineer, the IR2 who shall carry out an Inspection himself/herself and notify the RGE/RSE/ELS Designer / or ICE. Excavation work shall be temporarily suspended until IR2/ ELS Designer / or ICE has reviewed the situation or established remedial measures and passed this Hold Point after the remedial measures are completed to his/her satisfaction and a Hold Point Proforma Form C is signed off.
- vi. The above procedure is repeated until the final excavation level is reached.
- vii. A flow chart showing the procedures in this Inspection Plan is shown in **Appendix 1.**

Note: If the limiting “Alarm” level is reached at any time during the excavation, no further excavation shall proceed. The procedure as prescribed in the approved plan or construction drawing for such situation shall be followed.



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6. Communication, provision of Induction and Training for Inspection Procedure of Deep Excavation

- i) The Company's Chief Engineer shall communicate and formally issue the Quality Inspection/Supervision Plan for Deep Excavation to all construction projects for implementation.
- ii) The Company's Chief Engineer or his authorized deputy/representative will be responsible for the provision of training to enhance communication and awareness to all civil engineering, foundation and building projects which involve deep excavation on the specific requirements of Quality Inspection/Supervision Plan for Deep Excavation
- iii) When a contract is awarded or 3 months prior to commencement of the deep excavation, the Site Agent/Project Manager/Project Director shall arrange a date with the Company's Chief Engineer or his authorized deputy/representative for a quality induction of Quality Inspection/Supervision Plan for Deep Excavation. The induction shall be attended by designated IR2, IR1, Site Agent/Project Manager/Project Director.
- iv) Prior to deep excavation work, the designated IR2 shall arrange and provide a quality training to the designated IR1, frontline supervision staffs, sub-contractors' representative and relevant workers to ensure specific requirements of Quality Inspection/Supervision Plan for Deep Excavation are understood, timely communicated and effectively implemented. Further quality training to be provided at appropriate intervals as the designated IR2 considered to be required or when new workers joined the site.



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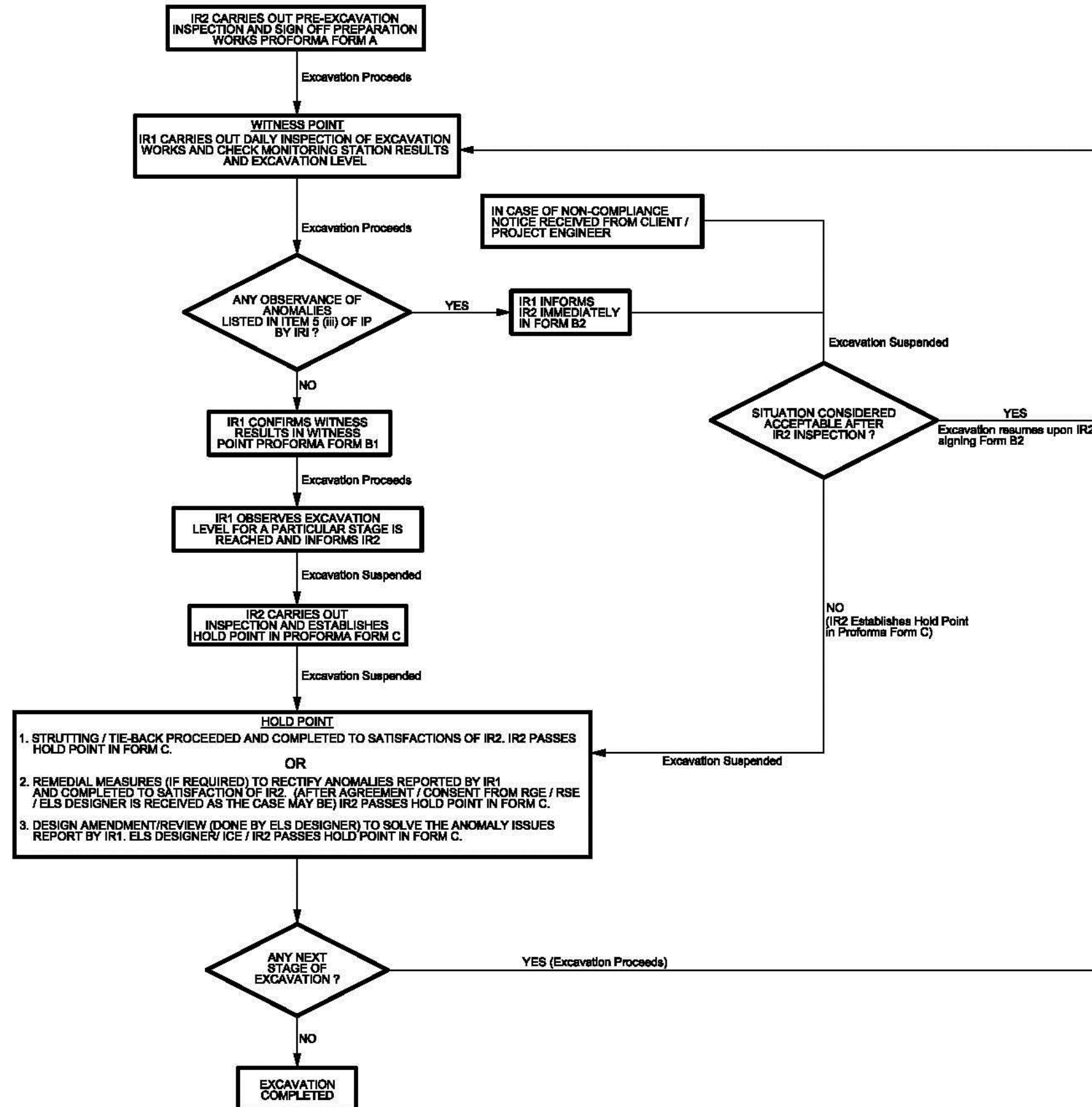
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APPENDIX 1
Flow Chart

FLOW CHART FOR QUALITY INSPECTION/ SUPERVISION PLAN FOR DEEP EXCAVATION

WITNESS POINT

HOLD POINT





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APPENDIX 2

Form A : Preparation Works Proforma

Ref. No. _____

Preparation Works Proforma

Project : _____

Type of Works : _____

Items for the Checklist of Pre-excavation Preparation Works

Item No.	Description
A1	Fence off works area by suitable means.
A2	Condition survey done.
A3	Monitoring stations / instrumentation as required in approved plans/construction plans installed. Initial monitoring data checked and submitted.
A4	Approved plans, SSP, GI Reports and temporary works design documents, approved method statements in place.
A5	Updated utility record plans checked and identified on site.
A6	De-watering equipment in place.
A7	Lighting and Ventilation equipment in place.
A8	Provisions for diversion of surface water into excavation done
A9	Check any Approval / Consent for commencement of Deep Excavation Works
A10	Implementation of effective communication channel within and between different supervision teams
*	Please refer to the note below

I confirm the above pre-excavation preparation works have been completed to my satisfaction and excavation may commence.

Signature :

Name of IR 2 :

Role/Post in this Project :

Date :

Time :

N.B. * Further items can be included to account for specific requirements of individual construction projects



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APPENDIX 3
Form B 1: Witness Point Proforma
With Annex 1

Ref. No. _____

Witness Point Proforma**Record of Inspection Performed by IR1****Project :** _____**Type of Works:** _____**Name :** _____**Grade of TCP :** _____ **Frequency of Inspection** daily

Date DD/MM/YY							
Item No.#	S/NS	S/NS	S/NS	S/NS	S/NS	S/NS	S/NS
1							
2							
3							
4							
5							
6							
7							
*							
Signature							

Legend S - Satisfactory

NS - Not satisfactory (if not satisfactory, complete **Form B2**)

- Descriptions of the items listed are attached in Annex 1

N.B. * Further items can be included to account for specific requirements of individual construction projects

Annex I to Form B1

Project : _____

Form B1 Ref. : _____

Witness Point Checklist for IR1

Item No.	Description
1	Construction Drawings / Approved Plans and method statement being followed.
2	Excavation level has not exceeded the designed level at this stage.
3	All monitoring results at monitoring check points below “Alert” level.
4	Lateral support works installed according to construction drawings/approved plans.
5	No abnormal movement or tilting of temporary works.
6	No unexpected inflow of water into excavation.
7	No damage report received to adjacent properties or utilities.
*	Please refer to the note below

N.B. * Further items can be included to account for specific requirements of individual construction projects



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APPENDIX 4

Form B 2: Non-conformity Report Proforma

Ref. No. _____

Non-Conformity Report**PART 1**

Project : _____

Record of Non- Conformity

Date discovered : _____

Details : _____

Non-conformity Reported to IR2 (Name of IR2) : _____

Time of Reporting : _____

Signature : _____

Name of IR 1 : _____

Date : _____

PART 2**Record of Inspection by IR2 / ELS Designer** (name)

Report received from :

On _____ (date) _____ (time)

Observation of Inspection : _____

Hold Point to be issued : *YES/NO Hold Point Form C Ref. No. _____

Excavation can resume : *YES/NO _____

Signature : _____ Signature : _____

Name of ELS Designer : _____ Name of IR 2 : _____

Company Name : _____ Role/Post in the Project : _____

Date : _____ Date : _____

Time : _____ Time : _____

* Delete if inapplicable _____



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APPENDIX 5
Form C: Hold Point Proforma

Ref. No. _____

Hold Point Proforma

Project : _____

Reason of Establishing this Hold Point:

* (i) Non-conformity Reported from IR 1 in Form B2

Ref. : _____

* (ii) Non-conformity Report received from Client / Project Engineer,

Ref.: _____

* (iii) A particular stage is reached where lateral support is to be installed.

*Remedial/Rectification work required: _____

Reason for Passing this Hold Point:

* Lateral support completed on :

* (i) above rectification work completed on :.

The undersigned certified that the *necessary rectification/remedial work to cover this non-conformity report/*lateral support has been completed satisfactorily and excavation work may resume.

Certified by IR2 : _____ Date : _____

Name: : _____

Role/Post in the Project : _____

ELS Designer : _____ Date : _____

Name : _____

Company Name : _____

* ICE : _____ Date : _____

Name : _____

Company Name : _____

* Delete if inapplicable